Splitters: a neck above every pin

East bay 2: a test of funnel splitters · 14 November 2026

The Galton board's pins are fair but not independent: a ball leaving a pin still moves sideways and goes the same way again two times in three (`FINDINGS.md`, section 6). The board's first pin is the exception: fed straight down a narrow neck, it splits the balls 50/50. Could a neck above every pin make every decision a fresh one? If so, a board of such splitters would show the mathematics of independent fair coins exactly, and with different walls it could show more than the binomial (gambler's ruin, first passage, the reflection principle). This is the test, in bpp, before any bay is designed around it.

The test bed

`splitter-test.lua`, run on its own (not in the hall). Rows of "teeth": each a 4 mm pin on top of a solid wedge whose sides slope down (60 degrees) into vertical necks; each neck drops a ball onto the pin below, half a pitch to the side, its foot 8 mm above that pin's centre (as the board's neck). The same balls (6 mm steel), plates (a 7 mm gap) and contacts (bounce 0.4, friction 0.3) as the Galton board. Balls are fed one at a time down a neck onto the first pin, with a random offset of up to 0.3 mm; each ball's path is recorded in the board's format and measured with `paths.py`. Eight rows.

```
SP_PATHS=/tmp/sp.txt SP_BALLS=600 bpp -f splitter-test.lua -n 24000
python3 paths.py /tmp/sp.txt
```

Sizes and contacts are set from the environment (`SP_S` the pitch, `SP_NECK_W`, `SP_NECK_L`, `SP_SLOPE`, `SP_BOUNCE`, `SP_FRICTION`, ...). Over 70,000 balls went through the plain necks in all the runs below; none stopped.

The first try: the memory moved into the neck

At a 20 mm pitch with necks 8 mm wide and 12 mm long, the balls went the same way 72% of the time, worse than the lattice. Logging how balls leave a neck (`SP_LOG=1`) showed why: a ball crosses the neck, meets the far wall and stays nearer it, leaving 0.4 mm off centre on the side it was heading to (78% of the time). The next pin sees it off-centre on that side.

But each part of the shape pushes the memory one way or the other (300 balls each; P(same) is the chance of going the same way as at the last pin, 0.5 for fair coins):

at a 20 mm pitch	P(same)	correlation 1 row apart
necks 8 wide, 12 long (the first try)	0.72	+0.44
necks 6.6 wide	0.41	-0.19
necks 7 wide	0.26	-0.49
necks 10 wide	0.80	+0.61
necks 4 long	0.99	+0.98
necks 24 long	0.34	-0.32
sides 45 degrees	0.72	+0.43

at a 20 mm pitch	P(same)	correlation 1 row apart
sides 75 degrees	0.29	-0.41
bounce 0.7	0.43	-0.15
bounce 0.1	1.00	+1.00
friction 0.05	0.48	-0.05
pitch 30 mm	0.52	+0.05

Narrow or long necks and steep sides make a ball turn back; wide or short necks and a dead ball make it go on. (It looks as if the number of times a ball crosses the neck decides the sign, odd or even; not checked.)

A 30 mm pitch: as good as independent, at its design point

The longer slopes bring a ball to the neck faster, to cross it more. With 1,000 balls each:

at a 30 mm pitch	P(same)	correlation 1, 2, 3 rows apart	spread (8 fair coins: 1.414)
necks 8 wide, 12 long	0.53	+0.07, +0.01, -0.02	1.47
necks 10 long	**0.49**	**-0.01, 0.00, +0.01**	**1.43**
necks 15 long	0.51	+0.01, +0.01, +0.01	1.46
necks 7.5 wide	0.50	+0.01, +0.04, 0.00	1.46
necks 8.5 wide	0.54	+0.08, +0.01, 0.00	1.49
sides 55 degrees	0.42	-0.17, +0.06, -0.03	1.26
sides 65 degrees	0.40	-0.20, +0.04, +0.01	1.21
friction 0.2	0.55	+0.09, +0.02, -0.02	1.55
friction 0.45	0.43	-0.15, +0.01, -0.01	1.24
bounce 0.3	0.69	+0.39, +0.18, +0.07	2.05
bounce 0.5	0.42	-0.16, +0.01, +0.02	1.24
pitch 40 mm	0.38	-0.24, +0.04, -0.02	1.11

(The correlations' uncertainty is about 0.01.) With necks 10 mm long the decisions cannot be told from fair, independent coins: P(same) 0.494, no correlation at any distance, the spread 1.43 against 1.414. That is the default of the test bed now.

- **Where the memory is small, it is one pin deep.** Two and three rows apart the correlations are within 0.06 in all these, against -0.21 on the lattice: what a splitter remembers is which way the ball came in. (Where the memory is strong, as at a bounce of 0.3 with short necks, it reaches two rows: below.)
- **The shape is forgiving:** a millimetre on the necks' width or length keeps P(same) within 0.49 to 0.54.
- **The bounce is not:** 0.3 gives 0.69, 0.5 gives 0.42. The slope (55 or 65 degrees) and the friction matter about as much.

Longer, narrower necks: less sensitive, but taller

With more crossings in a neck the memory should average out. P(same), 600 balls each:

	bounce 0.25	0.4	0.55	row to row
pitch 30, necks 8 wide, 12 long	0.69 (at 0.3)	0.53	0.42 (at 0.5)	39 mm
pitch 30, necks 8 wide, 24 long	0.66	0.55	0.63	51 mm
pitch 30, necks 8 wide, 36 long	0.59	0.47	0.65	63 mm
pitch 30, necks 7 wide, 24 long	0.64	0.48	0.64	51 mm
pitch 30, necks 7 wide, 36 long	**0.42**	**0.53**	**0.44**	63 mm
pitch 40, necks 8 wide, 24 long	0.54	0.61	0.69	59 mm
pitch 24, necks 8 wide, 36 long	0.78	0.38	0.53	54 mm

Over bounces of 0.25 to 0.55, the best (necks 7 mm wide and 36 mm long) stays within 0.42 to 0.53, against 0.42 to 0.69 for the short necks. It helps, but does not take the bounce out, and twelve rows of it would stand 0.76 m tall (the board's pins take 0.13 m).

Taking the bounce out: three tries

Mixing chambers. Partway down the neck it widens around a small post and narrows again: a little coin flip whose result is thrown away, mirror-symmetric so it cannot favour a side. Short chambers (4 mm) trap balls between the post and the narrowing wall. Longer ones pass them but are worse at a low bounce: at 0.25 the balls turned back 86 to 90% of the time. Every mixer in a vertical neck changes how many times a ball crosses, which is what the bounce already does.

Leaning necks. Lean the neck, and gravity holds a ball on its lower wall: it rolls out the same way whichever side it came in from, the bouncing damped instead of counted. The pin then goes where the ball lands, and the lean changes side row by row, so what one row favours the next undoes. It erased too much: the balls left the neck so alike that the pin became a knife-edge again (1 mm of the pin's place took a row from 98% one way to 96% the other), and at the balance point the little that still varied was the part that remembered: memory +0.75. A splitter needs plenty of randomness beside what it remembers; erasing the memory is no use if the randomness goes with it. The vertical neck's crossings are what give it that randomness, and why the bounce matters.

A corner. Building the wedges from two convex slices instead of one puts a seam where the slope meets the neck, at the corner a ball rolls over into the neck. Nothing else changes, but at a 30 mm pitch and 10 mm necks the memory moved from -0.01 to +0.12 (1,000 balls each, both reproducible to the ball). How a ball meets that corner matters that much. A printed corner (its radius, the layer lines, wear) will not be what any simulation assumed, so no simulation can say a printed splitter's memory to better than about 0.15.

(The memory here is `paths.py`'s row-by-row measure: P(right | right at the last pin) - P(right | left at it), averaged over the rows. It is 0 for independent pins even where a pin favours a side, which a pooled correlation would mistake for memory.)

Tuning instead: the necks' length for the bounce

So the splitters have to be tuned on the board that is built: necks as inserts in two or three lengths, swapped until the balls stop favouring "the same way" or "back". As a starting point, the memory and the spread (8 rows; fair coins 1.414) at a 30 mm pitch with necks 8 mm wide, 800 balls each (the memory's uncertainty about 0.03):

bounce	necks 6	8	10	12	14	17	20 mm
0.3	+0.56, 2.46	+0.57, 2.47	+0.39, 2.05	+0.38, 2.04	+0.37, 2.06	+0.17, 1.58	**+0.06, 1.56**
0.4	+0.36, 1.97	**+0.03, 1.44**	**-0.02, 1.41**	**+0.07, 1.46**	-0.20, 1.20	-0.09, 1.36	-0.27, 1.14
0.5	-0.30, 1.09	-0.44, 0.95	-0.33, 1.06	-0.16, 1.26	-0.12, 1.32	**0.00, 1.39**	**-0.02, 1.39**
0.6	-0.25, 1.08	-0.36, 1.02	**-0.03, 1.36**	+0.06, 1.53	+0.09, 1.58	+0.24, 1.79	+0.27, 1.82

Every bounce tried has a length near nothing, but they do not follow in order (0.4 wants 10 mm, 0.5 wants 17, 0.6 wants 10 again), and with the seam at the corner the table moves by up to 0.15 (at 0.3 no length came under +0.4). Where the memory is strong it reaches two rows (at 0.3 with 6 mm necks, +0.37 two rows apart). The table says where to start; the board says where to stop.

Teeth of one height: tuning without moving the rows

The table above changes the rows' spacing with the neck. A printed board needs its rows fixed, so the part to swap is the whole printed tooth (the wedge and its neck walls, keyed onto its steel pin), in variants of one height: 29 mm from the pin's centre to the necks' foot, the slope taking what the neck leaves (`SP_TOTAL=29.05`). Rows stay 37 mm apart whatever the teeth. Memory, spread (8 rows; fair 1.414) and the correlation two rows apart, 800 balls each:

bounce	neck 4 (66°)	6 (64°)	8 (62°)	10 (60°)	12 (57°)	14 (54°)	16 (50°)	18 (45°)	20 (39°)	22 (35°)	24 (30°)
0.3	+0.84, 3.23	+0.50, 2.19	+0.62, 2.52	+0.41, 2.02	+0.41, 2.01	+0.31, 1.95	+0.36, 1.93	+0.15, 1.65	**0.00, 1.50**	+0.36, 1.87	+0.37, 1.98
0.4	+0.53, 2.32	-0.24, 1.12	+0.09, 1.52	**+0.01, 1.48**	+0.11, 1.63	**+0.02, 1.46**	+0.22, 1.68	+0.24, 1.73	+0.13, 1.51	+0.12, 1.51	+0.06, 1.50
0.5	-0.29, 1.12	-0.32, 0.97	-0.26, 1.11	-0.30, 1.13	**-0.07, 1.37**	+0.09, 1.49	+0.07, 1.53	+0.28, 1.75			
0.6	-0.44, 0.93	-0.18, 1.23	-0.34, 0.99	**-0.04, 1.32**	+0.08, 1.57	+0.21, 1.61	+0.22, 1.74	+0.33, 1.87			

(Two rows apart the correlation stays under 0.08 wherever the memory is under 0.3.) The 12 mm tooth with friction 0.2 or 0.45 instead of 0.3, at a bounce of 0.4: -0.17 and +0.10; at 0.5: -0.08 and +0.05.

- **Every bounce has a tooth near nothing,** within 0.07: 0.3 the 20 mm neck, 0.4 the 10 mm (or 14), 0.5 the 12 mm, 0.6 the 10 mm.
- **Neighbours can differ wildly** (at 0.3: 20 mm 0.00, 22 mm +0.36), as the corner predicted. The tuning is by trial, not by reading between rows of the table.
- **The friction moves it about as much as a step of bounce,** so the choice is made on the board, not from the table.

So a splitter board comes with four sets of teeth: necks of 10, 12, 14 and 20 mm. Start from the measured bounce (the set the table marks), pour a few hundred balls, and compare the pile's spread with the binomial's (engraved in the bins): with a memory m the spread is about sqrt((1+m)/(1-m)) times the binomial's, so 10% too wide is about +0.1 and 10% too narrow about -0.1. A pile of 300 balls measures the spread to about 4%. Try the next set the table suggests, and keep the one closest.

A swallowing wall: first passage and the reflection principle

On a board, a swallowing wall is an exit in place of one column's necks: a ball reaching that column drops through a hole into its own tube, one tube for each row it can be caught at. Every pin a ball meets on the

way is an ordinary splitter. So the test is whether tuned splitters give the exact mathematics. `splitter-test.lua` takes `SP_WALL_L` and `SP_WALL_R` (places from the start; the row each ball is caught at, and the side, go to `SP_CAUGHT`), and `walls.py` compares the counts with the exact chances:

- a ball is first caught after n rows with probability (a/n) C(n, (n+a)/2) / 2^n, for a wall a places away (first passage);
- a survivor of all N rows ends at place u with probability [C(N, (N+u)/2) - C(N, (N+u+2a)/2)] / 2^N (the reflection principle: the paths that touched the wall pair off with the mirror images of paths ending at -2a-u).

Twelve rows, a wall 2 places to the left of the start, 3,000 balls, at a bounce of 0.4 (expected counts in brackets):

teeth	caught at row 2, 4, 6, 8, 10, 12	survivors at place 0, 2, 4, 6, 8, 10+	chi-square (11 df; 1% level 24.7)
tuned (10 mm necks, memory +0.01)	720, 368, 233, 176, 128, 86 (750, 375, 234, 164, 123, 97)	321, 402, 332, 168, 60, 6 (314, 419, 314, 152, 48, 10)	**11.6**
mistuned (4 mm necks, memory +0.53)	1182, 255, 160, 128, 109, 74	134, 193, 217, 182, 176, 190	4,354

With tuned teeth the board gives first passage and the reflection principle as the formulas do. Mistuned, it is plain at the first tube: the first two pins go together, so 1,182 balls are caught at row 2 instead of 750, and the survivors spread to the far side. No ball stopped.

For the exhibit, with 150 balls a run, a wall 2 places from the start fills the tubes with about 37, 19, 12, 8, 6 and 5 balls (a long thin tail: the average time to be caught is infinite), and about 63 balls survive, piled away from the wall (16, 21, 16, 8, 2). A wall 3 places away: 19, 14, 11, 8, 7, and 92 survivors.

Two swallowing walls: gambler's ruin

With a wall on each side, a ball leaves by the right one with probability a/(a+b) in the long run, for walls a places left and b places right of the start: a fair game cannot be beaten, so the chance is a straight line in where the ball starts. `walls.py` takes both walls and works the exact chances out row by row (for one wall they are the formulas above). Twelve rows, walls 6 places apart, tuned teeth, 2,000 balls each:

start (places from the left wall)	went right, of those caught (exact in 12 rows; long run)	caught left, right, survived (exact)	chi-square (12 df; 1% level 26.2)
1	0.135 (0.122; 0.167)	1545, 242, 213 (1548, 215, 237)	15.3
2	0.310 (0.297; 0.333)	1142, 513, 345 (1155, 489, 356)	7.8
3	0.506 (0.500; 0.500)	767, 787, 446 (763, 763, 475)	7.3

(The chi-squares are over the rows caught at on each side and the survivors' places.) Gambler's ruin as the mathematics has it. In all three slightly fewer balls survived than the exact count (1,004 against 1,068, about two standard deviations), as a slight memory left in the teeth would spread the balls a little faster; too little to show on a board.

For the exhibit the long-run line shows only if nearly every ball is caught. Walls 6 apart leave 12 to 24% of the balls alive after 12 rows; walls 4 apart only 2%, and the shares come to 0.246, 0.500 and 0.754 for starts 1, 2 and 3 places from the left wall (a quarter, a half, three quarters: about 37, 75 and 112 of 150

balls). So a gambler's-ruin board would have its walls 4 places (two pitches) apart, with the feed, or the walls, moved between runs.

The exhibit

`splitter-board.lua` builds the whole board on the Galton board's machine (README, "The splitter board"): twelve rows of splitters with 10 mm necks, and the walls as above, the exits now real (an exit block whose top slopes outward, a 6-degree ramp to a tube standing on the slide). Tour runs of 150 balls, each judged by a chi-square of the tubes and bins against the exact counts:

walls	result	chi-square (1% level)
first passage, three runs	caught 85, 85, 77; survived 65, 65, 73	4.3, 0.9, 4.2 (20.1)
none	the binomial	6.8 (16.8)
gambler's ruin, start 1	37 out right (37.5)	5.6 (18.5)
gambler's ruin, start 2	70 out right, 76 left (73, 73)	8.1 (20.1)
gambler's ruin, start 3	112 out right (112.5)	11.9 (18.5)

Two things the whole board needed that the test bed did not. A ball coming off a ramp meets its tube's outer wall and bounces back: with the wall between tubes only as high as the next tube's ramp, it bounced over into the next tube (the row 2 tube got 11 balls, the row 4 tube 39). Each tube's inner wall now rises to just under its own ramp's end. And with a wall an odd number of places from the start, the bins' dividers have to stand at the odd places, not counted from the wall (they stood in the middle of the bins, and balls came to rest on them).

What it means

- A neck above every pin can cut the memory to nothing, but only at a shape that suits the bounce, and how a ball meets the neck's mouth moves that shape. No passive mixer tried (chambers, a lean) took the bounce out.
- So a splitter board is to be tuned as built: four sets of teeth of one height (necks of 10, 12, 14 and 20 mm), the bounce measured first (`FINDINGS.md` section 5) to choose where to start. Watching a few hundred balls with `GB_PATHS`-style counting, or simply the spread of the pile against its engraved curve, says which way to go: too wide, the balls keep on (longer or shorter necks, by the table); too narrow, they turn back.
- With teeth of one height the rows are 37 mm apart whatever the set: twelve rows in 0.45 m.
- Nothing jammed in the plain necks (over 70,000 balls in all), though the balls came one at a time; a full feed is still to try.